

FAKE NEWS DETECTION USING MACHINE LEARNING AND DATA MINING

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1. Introduction

detecting fake news articles from textual content. The system aims to provide accurate predictions

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Augmented Reality (AR) interaction traditionally relies on large gestures, handheld controllers, or explicit voice commands, which create significant accessibility barriers for individuals with motor impairments. The uploaded documentation proposes a privacy-preserving, on-device continual multimodal micro-interaction recognition system designed to make AR usable through extremely subtle user actions such as micro-finger movements, gaze shifts, wrist motions, soft voice cues, or EMG muscle contractions.

The purpose of this system is to eliminate physical strain and cloud-based processing dependency by enabling all sensing, classification, and continual learning to run fully on-device. Current AR systems lack multimodal fusion, adaptive learning, and accessibility-centered thresholds—resulting in unreliable performance for users with fatigue, tremors, or inconsistent motor strength.

This research addresses these limitations by integrating:

- Multimodal sensing (camera, IMU, EMG, gaze, audio)
- On-device continual learning
- Adaptive physiological difficulty modeling
- Inclusive AR interface design

The goal is to create a low-fatigue, privacy-preserving, real-time AR interaction method tailored for accessibility.

Methodology

2.1 Purpose

The purpose of this project is to develop a machine learning-based framework capable of automatically

while remaining computationally efficient and suitable for deployment in real-world environments.

2.2 Research Design Approach

The project follows a supervised machine learning approach involving:

- Data collection from a publicly available dataset
- Data preprocessing and cleaning
- Feature extraction using TF-IDF
- Model training using Logistic Regression
- Performance evaluation using classification metrics

The overall workflow was designed to ensure effective handling of textual data and accurate binary classification.

2.3 Dataset Description

The dataset was obtained from the Mendeley Data Repository and consists of two separate files:

Fake.xlsx – Contains fake news articles

Real.xlsx – Contains authentic news articles

Dataset Statistics:

Fake News Records: 4,912

Real News Records: 5,294

Total Records: 10,206

Features per Dataset: 10 Columns

Classification Type: Binary Classification

Class Labels:

Real News = 0

Fake News = 1

The dataset exhibits a balanced distribution between fake and real news articles, reducing the likelihood of classification bias.

2.4 Development Tools and Technologies

The following tools and libraries were used:

- Python Programming Language
- Jupyter Notebook
- Pandas
- NumPy
- NLTK
- Scikit-learn
- Joblib
- Regular Expressions (re)

These technologies provide efficient support for data preprocessing, feature extraction, model training, and deployment.

2.5 Data Preprocessing

Text preprocessing plays a critical role in improving model performance. The following operations were applied:

1. Conversion of text into lowercase.
2. Removal of punctuation and special characters.
3. Tokenization of text.
4. Removal of stop words using NLTK.
5. Stemming using Porter Stemmer.
6. Elimination of irrelevant textual noise.

These preprocessing steps help standardize the dataset and reduce dimensionality.

2.6 Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was employed to transform textual data

into numerical vectors suitable for machine learning algorithms.

Advantages of TF-IDF:

- Identifies important words in documents.
- Reduces the influence of common terms.
- Produces sparse numerical representations.
- Improves classification effectiveness.

2.7 Machine Learning Model

Logistic Regression was selected as the classification algorithm because of:

- High efficiency in binary classification tasks.
- Fast training and prediction speed.
- Low computational requirements.
- Strong interpretability.
- Excellent performance on text classification problems

2.8 System Workflow

The proposed system follows the workflow below:

News Dataset

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Text Cleaning

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Tokenization

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Stopword Removal

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Stemming

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TF-IDF Vectorization

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Train-Test Split (80:20)

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Logistic Regression

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Prediction

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Fake / Real Classification

3. Theoretical Background

3.1 Fake News Detection

Fake news detection refers to the automatic identification of misleading or fabricated news content using computational techniques. Modern fake news detection systems employ machine learning algorithms to learn patterns from historical examples and predict the authenticity of unseen articles.

3.2 Natural Language Processing

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand and process human language. NLP techniques are used extensively in text classification, sentiment analysis, spam detection, and fake news detection.

3.3 TF-IDF

TF-IDF is a statistical method used to measure the importance of words within a document relative to an entire corpus. Words frequently occurring in a document but rarely appearing across all documents receive higher weights.

3.4 Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems. It estimates the probability that an input belongs to a particular class and generates predictions based on a decision threshold.

4. Experimental Results and Discussion

4.1 Dataset Split

The dataset was divided into:

Training Data: 80%

Testing Data: 20%

Training Samples $\approx$ 8,180

Testing Samples = 2,026

4.2 Classification Performance

Classification Report:

Class 0 (Real News)

Precision = 0.95

Recall = 0.97

F1-Score = 0.96

Support = 1054

Class 1 (Fake News)

Precision = 0.97

Recall = 0.94

F1-Score = 0.96

Support = 972

Overall Accuracy = 95.95%

Macro Average F1-Score = 96%

Weighted Average F1-Score = 96%

Class	Precision	Recall	F1-Score	Support
Fake News (0)	0.95	0.97	0.96	1054
Real News (1)	0.97	0.94	0.96	972
Accuracy	-	-	0.96	2026
Macro Average	0.96	0.96	0.96	2026
Weighted Average	0.96	0.96	0.96	2026

Model Accuracy: 95.95%

The Logistic Regression model achieved an accuracy of 95.95% on the test dataset. The model demonstrated strong performance in distinguishing between fake and real news articles, with both classes achieving an F1-score of 0.96. The high precision and recall values indicate that the model effectively minimizes both false positives and false negatives, making it a reliable solution for automated fake news detection.

4.3 Confusion Matrix Analysis

Confusion Matrix:

	Predicted Real	Predicted Fake
Actual Real	1027	27
Actual Fake	55	917

Interpretation:

Correctly Classified Real News = 1027

Correctly Classified Fake News = 917

False Positives = 27

False Negatives = 55

The low number of classification errors demonstrates the robustness of the proposed model.

4.4 Discussion

The Logistic Regression model achieved an overall accuracy of 95.95%, indicating excellent capability in distinguishing fake and real news articles.

The high precision values suggest that the model produces very few false classifications, while the strong recall values indicate effective detection of both fake and real news content. These results demonstrate that traditional machine learning approaches remain highly competitive when combined with appropriate text preprocessing and feature engineering techniques.

5. Research Gap and Contributions

5.1 Research Gap

Many existing fake news detection systems rely on deep learning architectures such as CNNs, LSTMs, and Transformers. Although these methods often achieve high accuracy, they require extensive computational resources and large training datasets.

There is limited research investigating whether lightweight machine learning approaches can achieve comparable performance on multilingual translated news datasets.

5.2 Contributions

The key contributions of this project include:

- Development of an automated fake news detection framework.
- Utilization of a multilingual translated dataset.
- Implementation of NLP-based preprocessing techniques.
- TF-IDF feature engineering.
- Achievement of 95.95% classification accuracy.
- Creation of a deployable machine learning pipeline using Joblib.

6. Applications

The proposed system can be applied in:

Social media monitoring

- News verification platforms
- Online journalism
- Government information systems
- Election misinformation detection
- Fact-checking organizations
- Educational content verification

7. Conclusions

This project successfully developed a machine learning-based fake news detection system using Data Mining and Natural Language Processing techniques. The combination of TF-IDF feature extraction and Logistic Regression classification achieved an overall accuracy of 95.95% with balanced precision, recall, and F1-scores.

The results demonstrate that lightweight machine learning approaches remain highly effective for fake news detection while offering lower computational

complexity compared to many deep learning alternatives. The developed framework provides a practical and scalable solution for combating misinformation in digital environments.

Future work may involve integrating Transformer-based architectures such as BERT, supporting multilingual native-language datasets, implementing explainable AI techniques, and deploying the system as a real-time web-based fact-checking platform.

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